Audio Programming

“Petrichor” - Generative Drone Synthesis

1. Introduction

Petrichor is a generative drone instrument exploring sustained harmonic colour, subtle spectral motion, and long-form structural development. Rather than relying on melody or rhythm, the piece evolves through gradual timbral transformation and layered modulation processes. The title refers to the scent of earth after rainfall — an atmospheric presence that emerges slowly and dissipates gently. This idea informs both the sonic material and the formal trajectory of the work.

The piece unfolds across three stages: gradual emergence, extended suspended texture, and controlled dissolution. The system is designed to develop independently once initiated, allowing internal processes to shape the musical form over time.

2. Ideation and Development

The ideation process began with the question of how to maintain listener engagement without rhythmic or melodic events. Early experiments with static oscillators revealed that sustained tones alone quickly become inert. As a result, the focus shifted toward internal motion — using modulation, filtering, and spatial movement to create evolving texture.

Extended chord voicings were selected to provide harmonic richness without frequent change. Rather than fixed timing, chord transitions were assigned irregular durations to avoid predictable looping. The aim was to simulate organic drift rather than mechanical sequencing.

Additional layers were introduced gradually: amplitude modulation to create a breathing quality, spectral emphasis to introduce shimmer, and a subtle atmospheric texture component to soften the purity of the sine-based material. Each element was chosen to enhance depth while preserving restraint.

3. Sonic Structure and Analysis

The overall signal flow is shown below.

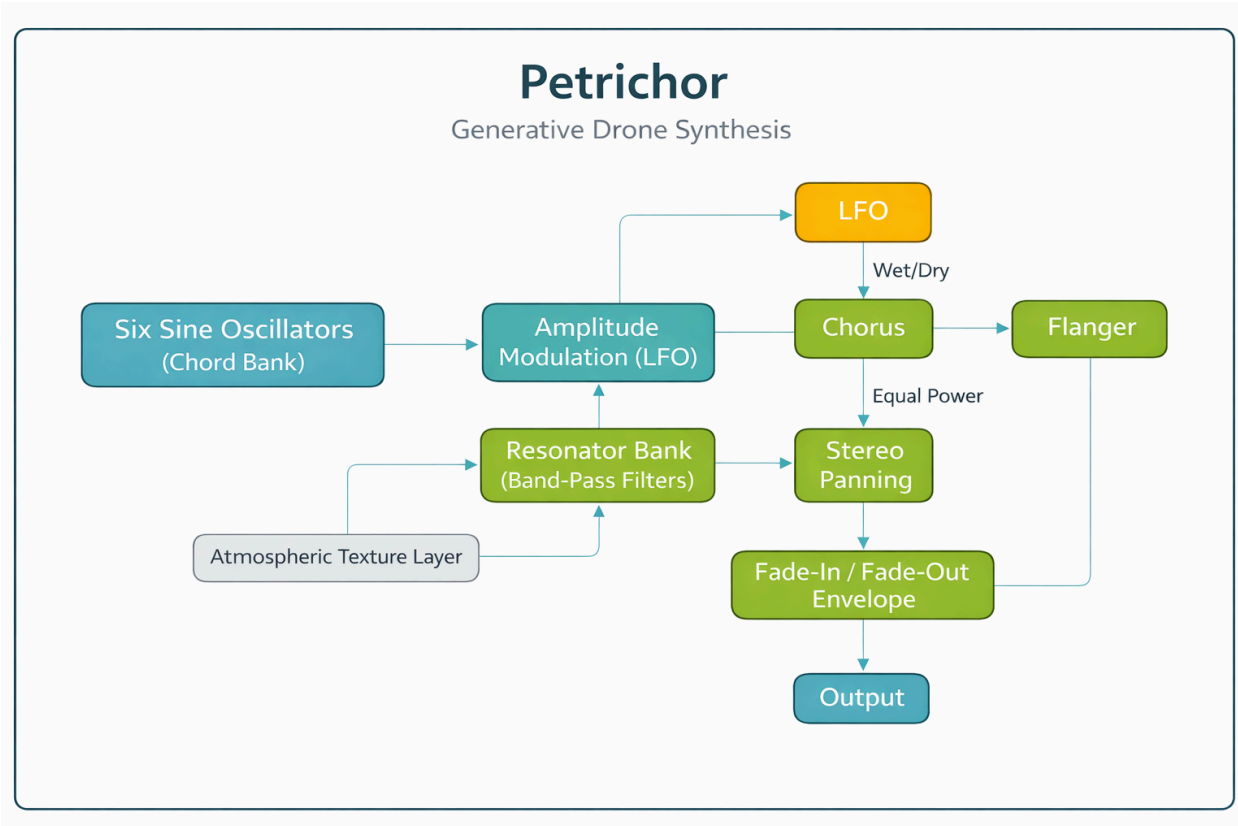

Figure 1: Simplified signal flow of the Petrichor generative drone instrument.

The system begins with a six-voice sine oscillator bank forming extended harmonic fields. Two related chord states alternate over long durations, creating slow harmonic evolution.

Amplitude modulation introduces gentle dynamic movement. A resonator bank of band-pass filters enhances upper partials, producing shimmer and fine spectral detail. An atmospheric texture layer contributes diffuse grain, preventing the harmonic body from sounding overly pure.

Stereo panning is implemented using equal-power distribution, allowing gradual spatial drift. Chorus and flanger stages introduce detuning and comb-filter motion, increasing depth and spectral animation. A slow modulation of wet/dry balance ensures continuous yet understated variation.

Finally, fade-in and fade-out envelopes shape the form. The curved fade-out provides intentional closure, allowing the texture to dissolve naturally.

Through layered modulation, spectral shaping, and long-duration timing structures, *Petrichor* demonstrates how minimal harmonic material can sustain attention through slow, continuous transformation.